

UNIT 3: CELLS

3.3 – CHROMOSOMES

Biology 9 Mr. Queenan
Ch. 15.1 (p.474-479)

Questions?

- What is a chromosome?
- Where in the cell is it located?
- What is the importance of a chromosome?
- Are chromosomes the same in all organisms?

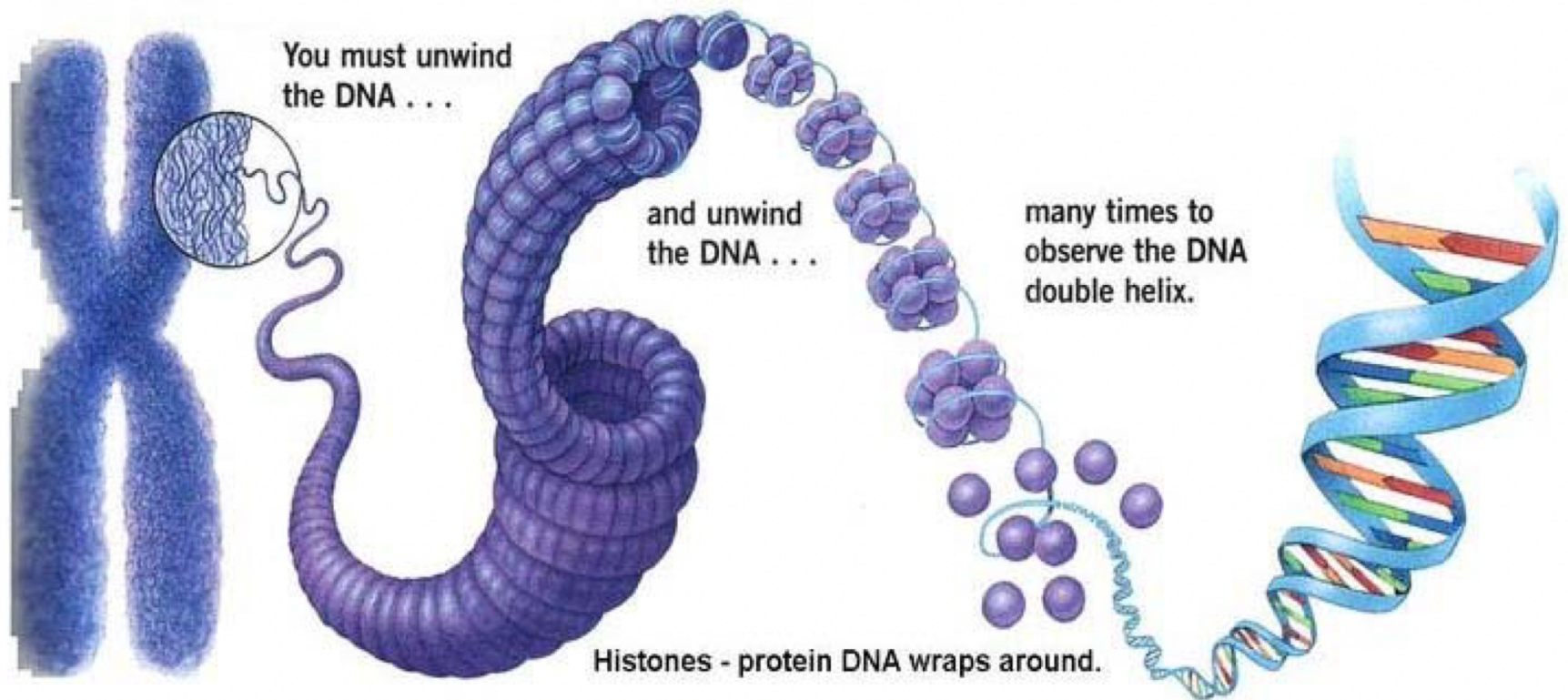
Chromosome Structure

- DNA in a human cell contains ~ **3 billion nucleotides**
 - ▣ Approx. size of a **basketball**
- DNA must be **coiled** and **packaged** in such a way that:
 - ▣ It **fits** within the **nucleus**
 - ▣ It can be **further compacted** during **cell division**
 - ▣ It can be **unpacked** for **transcription**
- During **cell division** DNA is **coiled** into **compact structures: chromosomes**
 - ▣ **Rod-shaped** structures of **DNA** and **proteins**
 - ▣ Contain **multiple layers of coiling/compaction**



Chromosome Structure

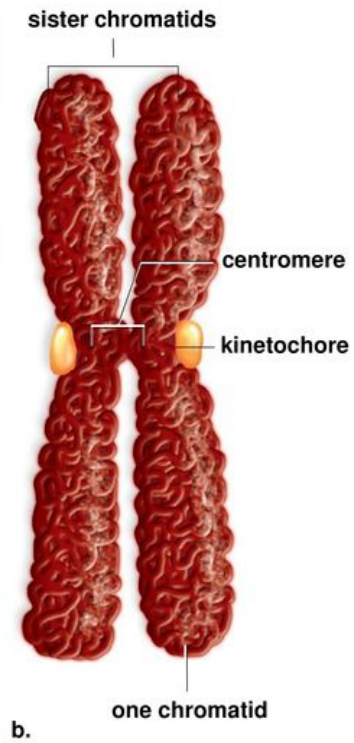
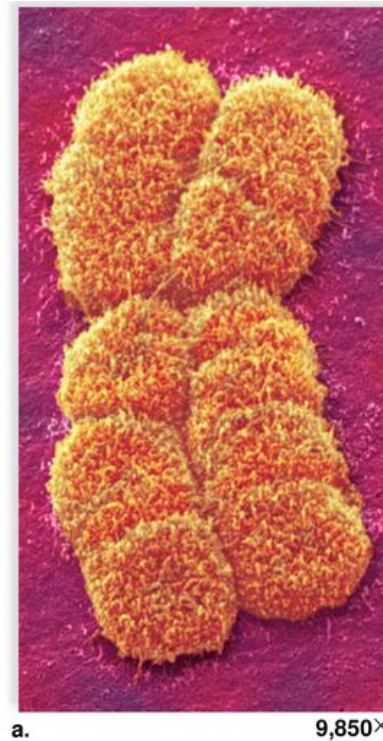
- Each **chromosome** contains **single DNA molecule wrapped** tightly around **histone proteins**
 - ▣ **Histone** – helps maintain shape of chromosome and aid in packing
 - ▣ **Other proteins associated** with **DNA** involved in controlling activity of specific regions
- **Between cell division** DNA is **less tightly packed**
 - ▣ **Regions uncoil** between divisions in order to **guide activities of cell (transcription/translation)**
 - ▣ **Less packed** DNA known as **chromatin**



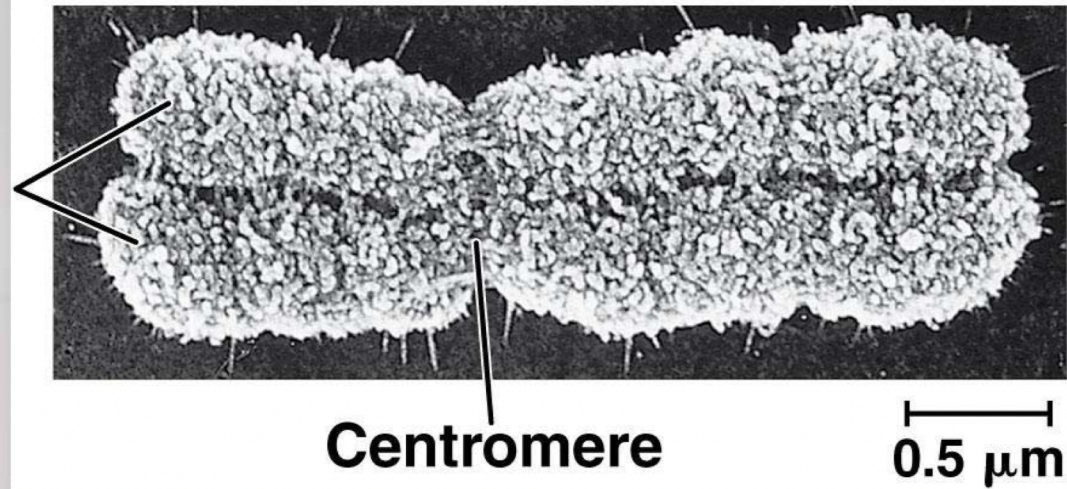
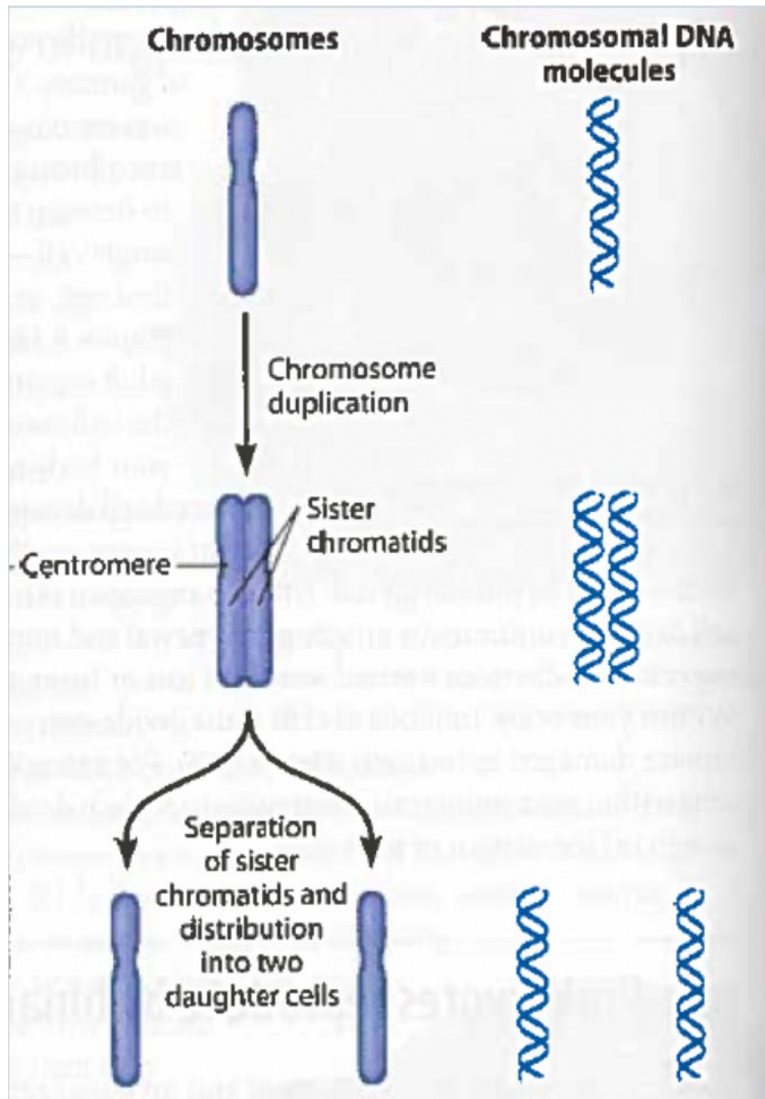
Histones

Chromosome Structure

- Replicated chromosomes consist of **2 identical halves**
 - ▣ Each half: **chromatid**
 - ▣ Chromatids **form** as **DNA copies itself** prior to **cell division**
 - ▣ As cell divides, **each new cell** gets **1 chromatid** (1 copy of **each chromosome**)
- Chromatids **attach** at the **centromere**
 - ▣ **Holds** the **chromatids together** until they **separate during division**
 - ▣ Important for **movement of chromosomes** during **division**



Chromosome Structure

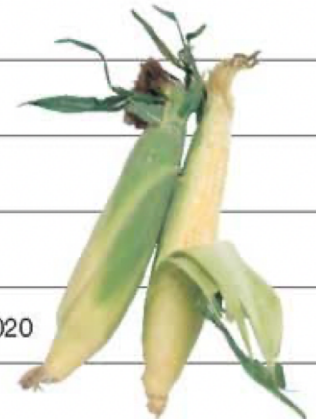
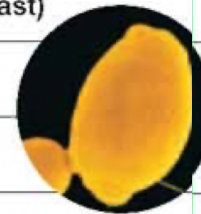


Chromosome Number

- **Every species has characteristic number of chromosomes per cell; some can have same number**
- ▣ **Number of chromosomes/cell does **not** necessarily correlate with organism complexity**
 - Some plants and simple multicellular organisms have more chromosomes than humans

Rank the following organisms 1-11 by chromosome # (lowest to highest)

Organism	Number (2n) of chromosomes
<i>Penicillium</i>	1-4
<i>Saccharomyces</i> (yeast)	16
Mosquito	6
Housefly	12
Garden pea	14
Corn	20
Adder's tongue fern	480-1,020
Frog	26
Human	46
Orangutan	48
Dog	78

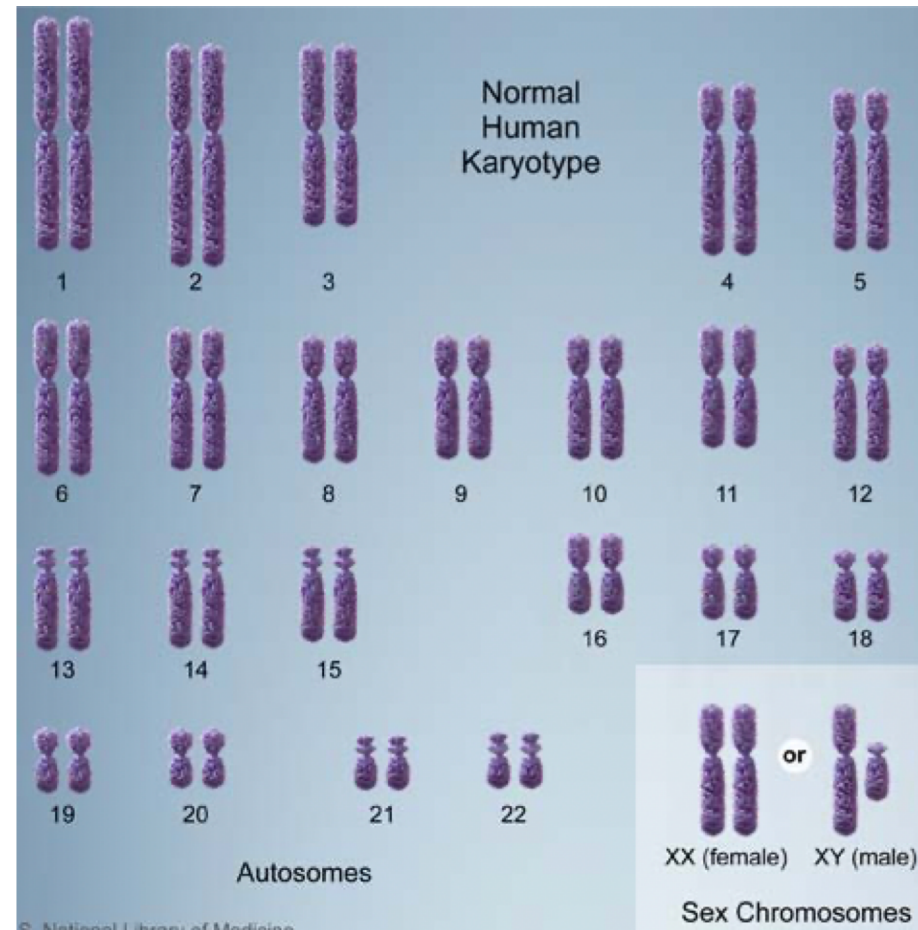


Chromosome Number

- **Chromosomes** categorized as **autosomal** or **sex** chromosomes
 - ▣ **Sex chromosomes determine sex of organism; carry some other characteristics**
 - In **humans**, known as **X and Y** chromosomes
 - Males: **XY**; females: **XX**
 - ▣ **All other** chromosomes known as **autosomal**
- **Humans** have total **46 chromosomes**
 - ▣ **2 sex** chromosomes
 - ▣ **44 autosomal** chromosomes

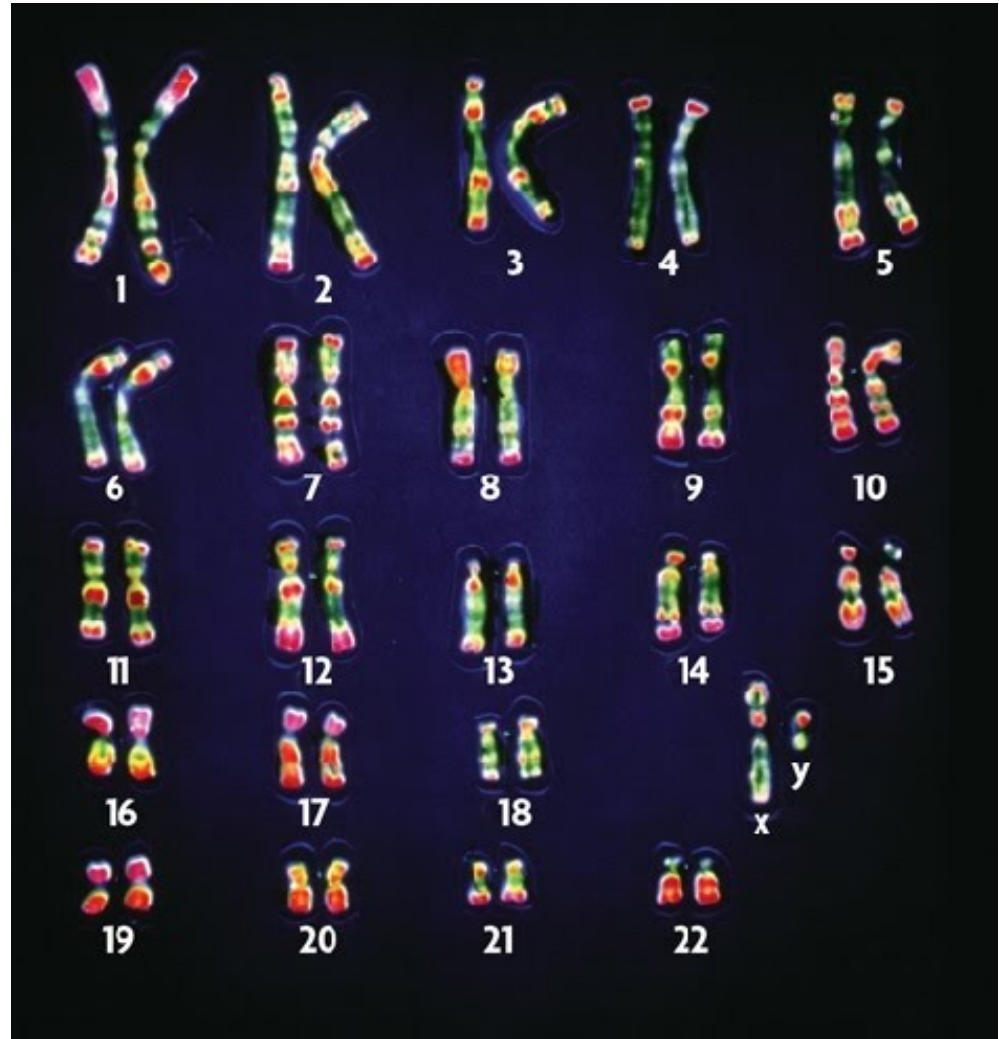
Homologous Chromosomes

- **Each cell** of an organism produced by **sexual reproduction** has **2 copies** of each **autosome**
 - ▣ **One copy** received from **each parent**
- The **2 copies** known as **homologous chromosomes** (**homologues**)
 - ▣ Same **size, shape**, carry **same genes** for **same traits**



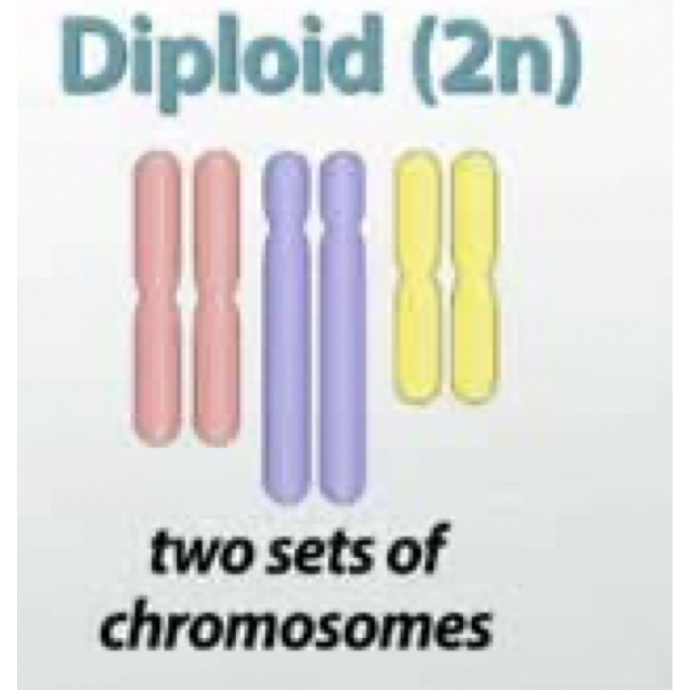
Homologous Chromosomes

- **Karyotype:**
photomicrograph
of chromosomes in
a dividing cell
- **46 human**
chromosomes:
 - ▣ **22 homologous**
pairs of autosomes
 - ▣ **2 sex chromosomes**



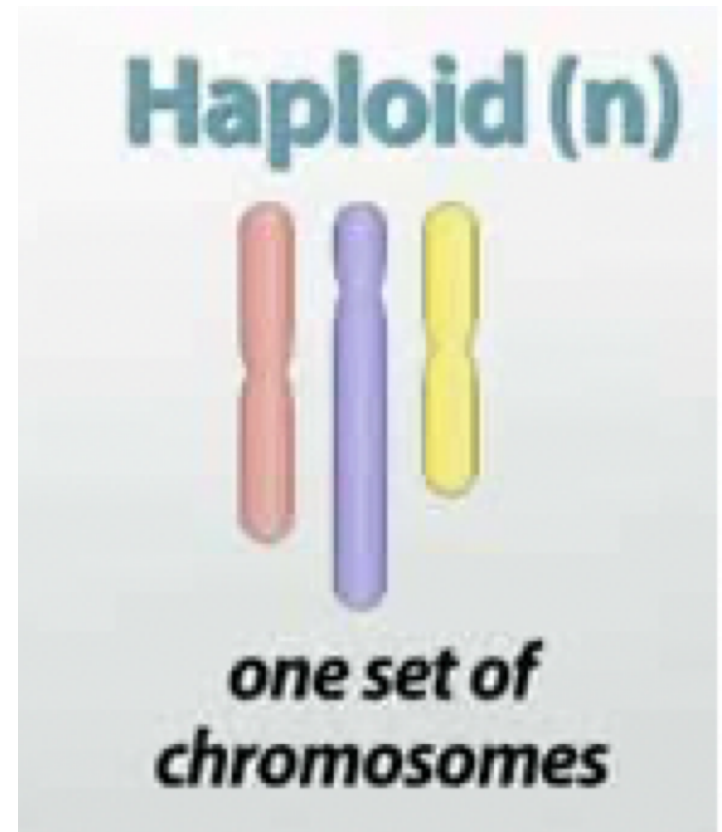
Haploid vs Diploid Cells

- Cells that have **2 sets of chromosomes** are **diploid**
 - ▣ **2 autosomes** for each homologous pairs
 - ▣ **2 sex** chromosomes
 - ▣ **All human cells** are **diploid**
 - **Except reproductive cells** (sperm/egg)
- **Diploid** abbreviated as **2n**
 - ▣ **Human diploid (2n) number = 46**
 - ▣ **2 x 22 autosomes + 2 sex** chromosomes



Haploid vs. Diploid Cells

- **Reproductive cells are haploid** – contain only **one set of chromosomes**
 - ▣ **Half number of chromosomes in diploid cells**
 - ▣ **One autosome from each homologous pair and one sex chromosome**
 - **22 autosomes + 1 sex chromosome = 23 total**
- **Haploid abbreviated as $1n$**
 - ▣ **Sperm cell ($1n$) combines with egg cell ($1n$) to create new cells of new organism ($2n$)**



Prokaryotic Chromosomes

- **Prokaryotic chromosome much simpler**
 - ▣ **Single, circular chromosome attached to inside of cell membrane**
 - **Also compacted to fit within cell**
 - ▣ **Region known as the nucleoid**
- **Between cell divisions: DNA less compacted as chromatin**
- **In preparation for/during cell division: DNA highly compacted to form chromosomes**

